

Advanced Computer Networks Lab



Dr Sudipta Saha

Associate Professor

Dept of Computer Science & Engineering

Indian Institute of Technology Bhubaneswar

Lab Assignment - 3.1-3.2

Teaching Assistants: Satwik and Arindam



**DSSRG: Decentralized
Smart Systems Research
Group**

<https://sites.google.com/iitbbs.ac.in/dssrg>

Or Google dssrg iitbbs



About

- Transport layer protocols: TCP vs UDP
- TCP is connection oriented, ensures data integrity through ordering, acknowledgement and retransmission.
- UDP is fast, connectionless, and unreliable protocol.
- TCP has everything, why UDP then?
- UDP gives blank slate. Implement additional functionality as required.
- This assignment focuses on such applications that can be created on top of UDP.



Learning outcome (A3.1 and 3.2)

- How to transfer binary data via sockets.
- Parsing binary from sockets to data structure.
- Video streaming application on top of UDP.
- How basic video streaming works.
- How to efficiently transfer data (video frames) over network.
- Make the connection reliable.



Assignment 3.1

Problem Statement:

Build a simple video streaming client for a video streaming server provided to you in executable form. The video streaming would be built on top of UDP. After running the server it will wait for client to get connected. The client when connected will first receive video file metadata, then it will receive all the frames. The frames would be sent one after another automatically, the client just have to read the frames and display them. You don't need to implement the video player GUI, you have to use the OpenCV library.

Download sample video from:

<https://www.peach.themazzone.com/durian/movies/sintel-1024-surround.mp4>



Assignment 3.1



The screenshot shows a Windows terminal window with a dark background. The title bar at the top indicates the user is 'dell@DESKTOP-09BUBKJ' in the directory '/mr'. The terminal content shows the user running the command 'build2/server' in the directory '/mnt/d/projects_wsl/acnlab-udp-video-streaming'. The output shows 'Video streaming server' and 'server: waiting to recvfrom...'. The status bar at the bottom is green and displays '[0] 0:bash*' on the left and '"DESKTOP-09BUBKJ" 15:08 20-Mar-26' on the right.

```
dell@DESKTOP-09BUBKJ: /mr × Settings × + v
dell@DESKTOP-09BUBKJ:/mnt/d/projects_wsl/acnlab-udp-video-streaming$ build2/server
Video streaming server
server: waiting to recvfrom...
```

[0] 0:bash* "DESKTOP-09BUBKJ" 15:08 20-Mar-26

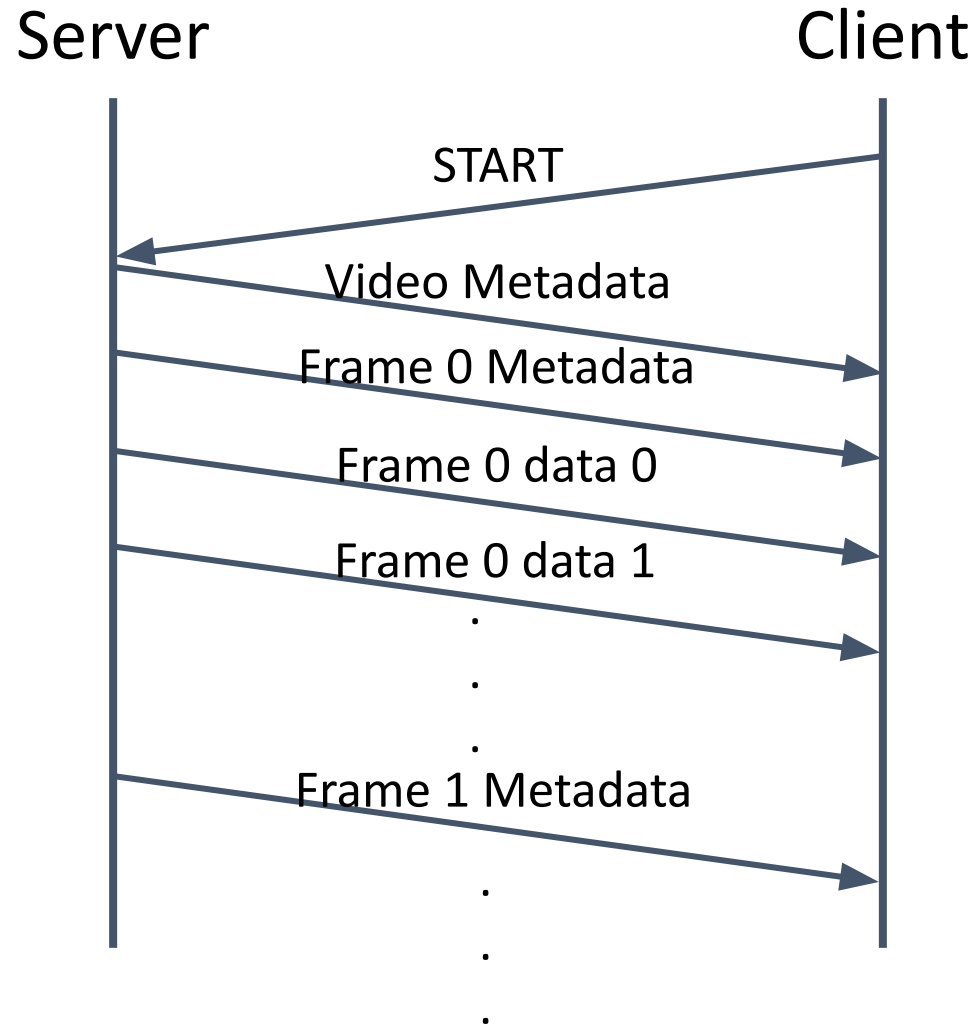


Assignment 3.1

A screenshot of a Blender 2.80.0 interface. The central viewport displays a video player window titled "sintel-1024-surround.mp4". The video content shows a title card for "THE BLENDER FOUNDATION PRESENTS" over a snowy, mountainous landscape. The left sidebar contains a list of frames from 103 to 135. The right sidebar shows a list of client messages, including "client: displaying image" and "client: frame135". The bottom status bar shows the path "0:build2/client*" and the system time "15:13 20-Mar-26".



Assignment 3.1



Assignment 3.1

updated slide

Video metadata

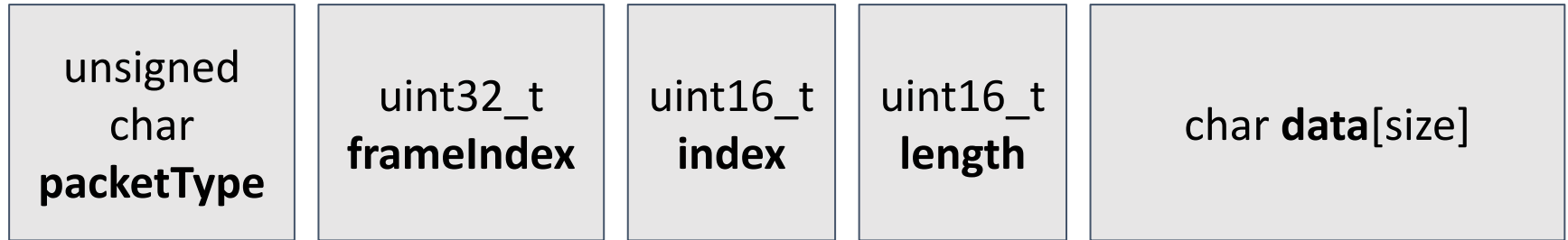
unsigned char packetType	char name[1024]	uint8_t fps	uint16_t height	uint16_t width	uint32_t frameCount
---------------------------------------	---------------------------	-----------------------	---------------------------	--------------------------	-------------------------------

Frame metadata

unsigned char packetType	uint32_t index	uint16_t dataPacketCount	uint32_t size
---------------------------------------	--------------------------	------------------------------------	-------------------------

Assignment 3.1

Frame data packet



*#define MAXBUFLLEN 32 * 1024*

size = MAXBUFLLEN - extra_size

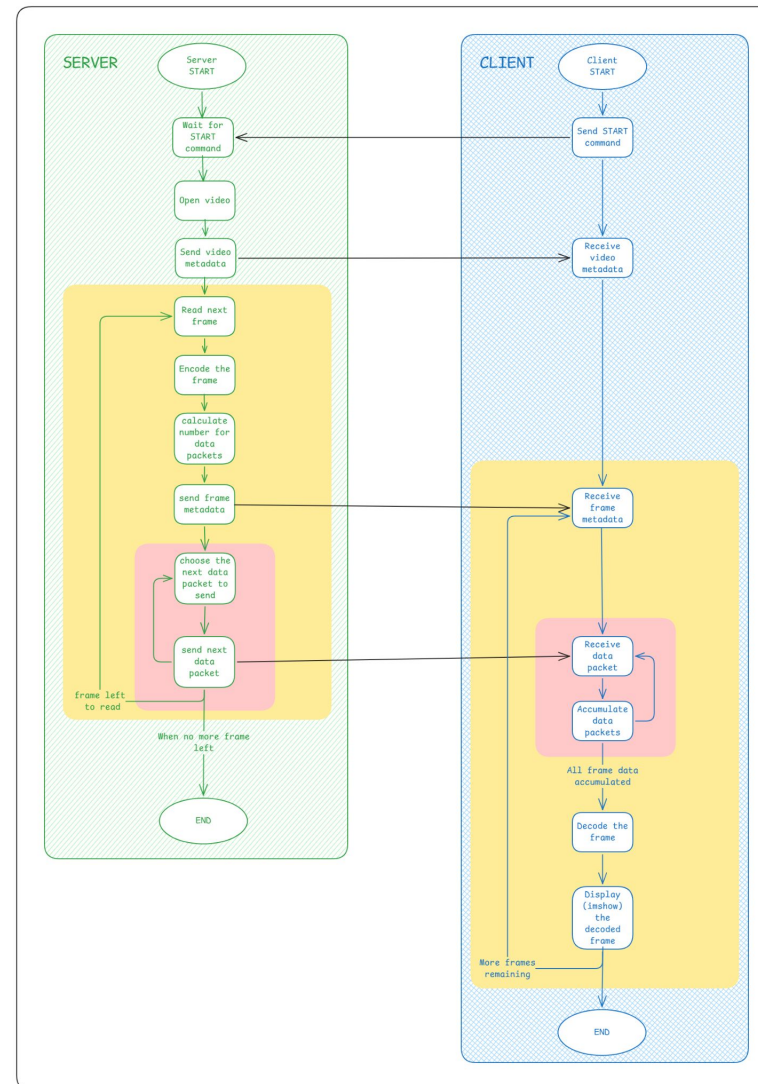
*extra_size = sizeof(unsigned char) + sizeof(uint32_t)
+ 2 * sizeof(uint16_t)*

Assignment 3.1

packetType:

- 1 - Video metadata
- 2 - Frame metadata
- 3 - Frame data packet

Assignment 3.1



Assignment 3.1

```
dell@DESKTOP-09BUBKJ:/mnt/d/projects_wsl/acnlab-udp-reliable-video-streaming$ build/server -h
Video streaming server
udp video streaming server

server [OPTIONS]

OPTIONS:
-h,      --help                Print this help message and exit
-f,      --filename TEXT [sintel-1024-surround.mp4]
                               Video filename path
-p,      --port TEXT [8000]    Port of the server
-d,      --delay-between-frames INT [30]
                               Delay between sending frames (>0)
-q,      --encode-jpeg-quality INT [80]
                               JPEG quality while encoding each frames (0-100)
```



Assignment 3.1

Run the server:

- `build/server` will start the server with port 8000 and read and send the video sintel-1024-surround.mp4
- `build/server -p 8001` will start the server in different port (here 8001)
- `build/server -f [video_path]` will read a different video



Assignment 3.2

Problem Statement:

In previous assignment 2.1, you implemented a simple video streaming client. The channel was assumed to be noiseless, no packet would be dropped and they will be received in sequence. In assignment 2.2, you have to implement the video streaming client making sure that the video frames are received in order and dropped packets should be re-requested after a certain duration. Again the video streaming server would be provided in form of executable and this time you have to request for a frame rather than server sending them automatically.

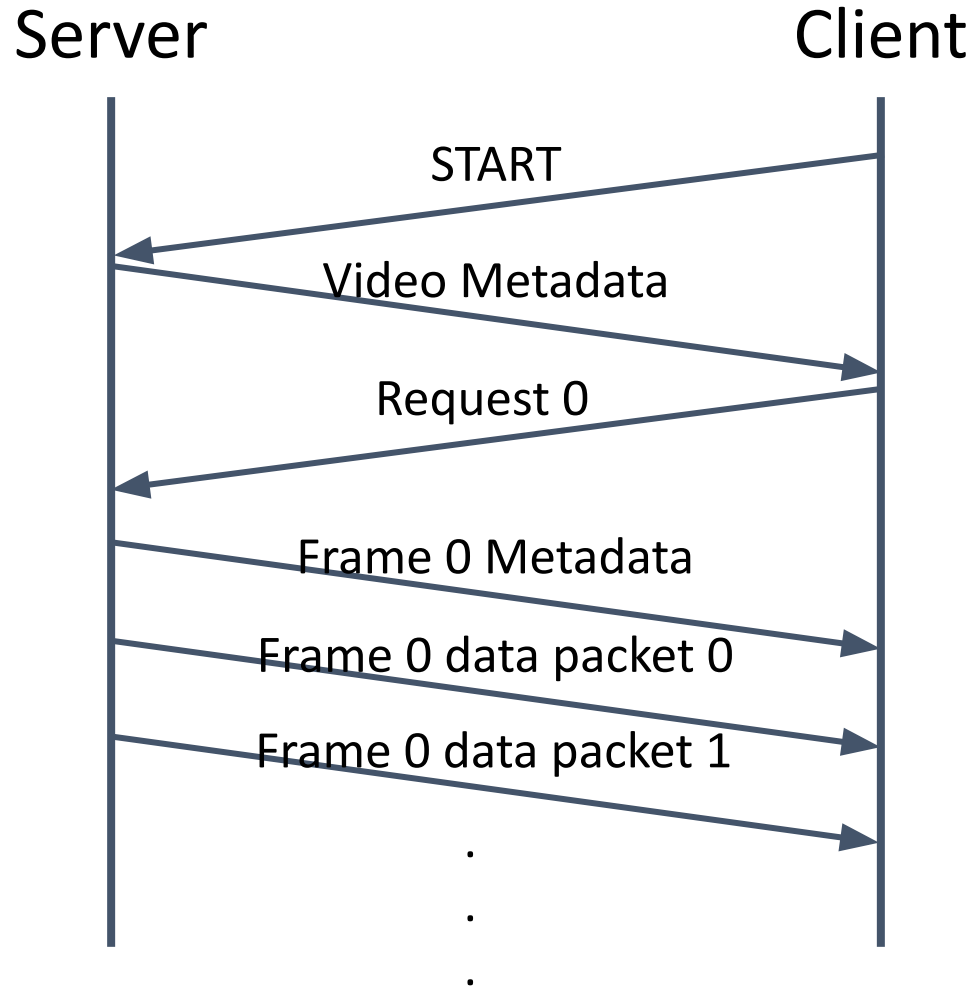


Assignment 3.2

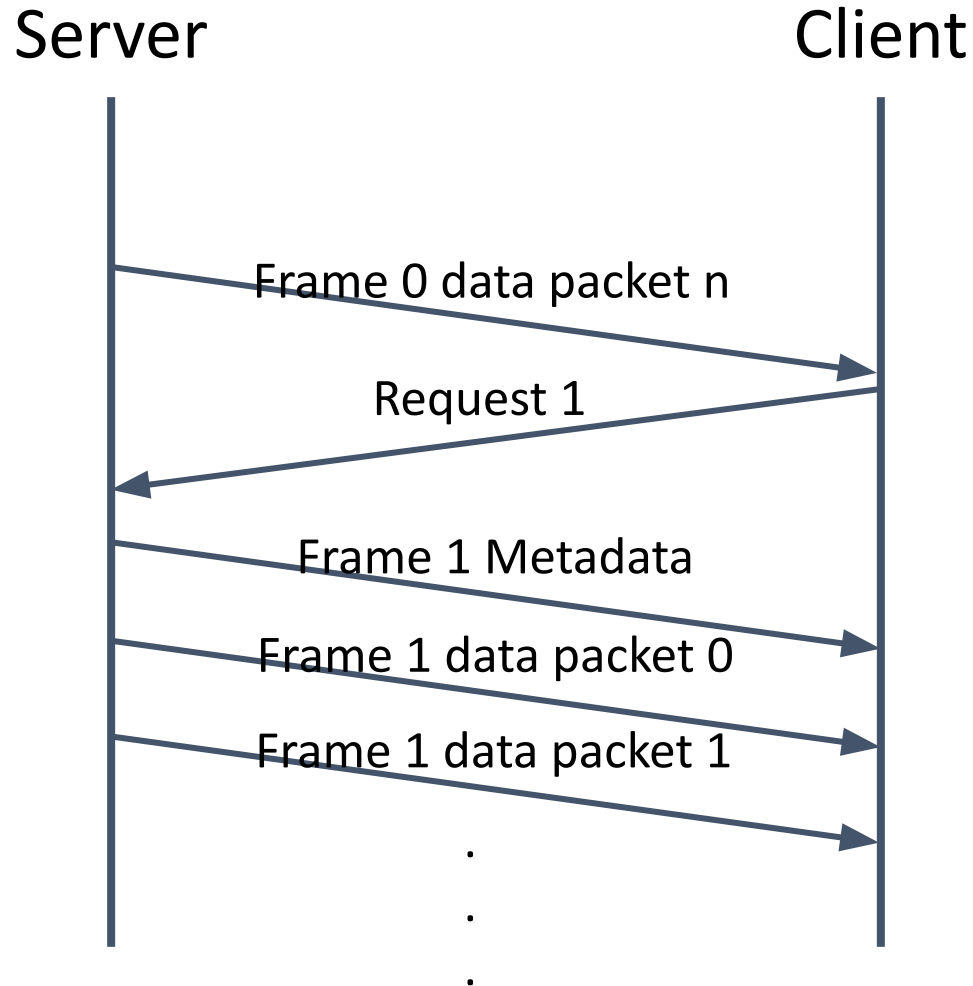
The image is a screenshot of a computer screen displaying the Blender 2.80.0 interface. The top of the screen shows the Blender window title bar with the text "dell@DESKTOP-09BUBKJ: /mr" and standard window controls. Below this, the main 3D viewport is visible, showing a scene with a light blue sky and a dark, silhouetted landscape. In the center of the viewport, the text "THE BLENDER FOUNDATION PRESENTS" is displayed in a serif font. Overlaid on the bottom right of the 3D viewport is a smaller window titled "sintel-1024-surround.mp4", which shows a video frame of a snowy landscape with a small figure in the distance. At the bottom of the screen, a terminal window is open, displaying a list of frame numbers from "frame103" to "frame135" on the left, and a series of log messages "client: displaying image" and "client: frame135" on the right. The terminal prompt is "[0] 0:build2/client*". The bottom right corner of the screen shows a system tray with the text "DESKTOP-09BUBKJ" 15:13 20-Mar-26.



Assignment 3.2



Assignment 3.2





Assignment 3.2

additional slide

packetType:

- 1 - Video metadata
- 2 - Frame metadata
- 3 - Frame data packet
- 4 - Request frame



Assignment 3.2

additional slide

Request frame

unsigned
char
packetType

uint32_t
frameIndex



Assignment 3.2

```
dell@DESKTOP-09BUBKJ:/mnt/d/projects_wsl/acnlab-udp-reliable-video-streaming$ build/server2 -h
Video streaming server
udp video streaming server
```

server [OPTIONS]

OPTIONS:

```
-h,      --help                Print this help message and exit
-f,      --filename TEXT [sintel-1024-surround.mp4]
                               Video filename path
-p,      --port TEXT [8000]    Port of the server
-d,      --delay-between-frames INT [30]
                               Delay between sending frames (>0)
                               --packet-drop-prob FLOAT [0.1]
                               Packet drop probability (0-1)
-q,      --encode-jpeg-quality INT [80]
                               JPEG quality while encoding each frames (0-100)
-r,      --disable-random-packet-drop
                               Enable random packet drop
-s,      --fixed-seed          Fixed seed for random number generator
-c,      --continuous-packet-drop
                               Drop packet throughout the runtime.
```



Assignment 3.2

Run the server:

- `build/server2` will start the server with port 8000 and read and send the video sintel-1024-surround.mp4
- `build/server2 -p 8001` will start the server in different port (here 8001)
- `build/server2 -f [video_path]` will read a different video
- `build/server2 -r` to disable random packet drops
- `build/server2 -packet-drop-prob 0.3` to set the packet drop probability to 0.3



Thank you

